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MILITARY SPECIFICATION

TEST SET, ARMAMENT WIRING SYSTEM,, SPECIAL WEAPON AN/GWM-8A

This specification is approved for use by SA-ALC/SWRCS, Kelly AFB TX 78241, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for one type of Test Set, Armament Wiring System, Special Weapon AN/GWM-8A.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-4158	Electronic Equipment, Ground, General Requirements for
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-C-25200	Cable Assemblies, Special Weapons, Electrical, General Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/SWRCS, Kelly AFB TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1190

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STANDARDS

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-810	Environmental Test Methods

DRAWINGS

Air Force

59D45036	Test Set, Armament Wiring System, Special Weapon AN/GWM-8A, Assembly of
59C45037	Cable Assembly, Special Purpose, Electrical CX-4701/GWM-8A
59C45039	Wiring Schematic

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute Inc

ANSI Y32.16	Reference Designations for Electrical and Electronic Parts and Equipments

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York NY 10018.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provisions for preproduction testing.

3.2 General. The requirements of MIL-E-4158 and MIL-C-25200 are applicable as requirements of this specification. When conflict occurs with this specification, the requirements of this specification shall govern.

3.3 Components. The Set, Armament Wiring System, Special Weapon AN/GWM-8A shall consist of the following:

a. The testing unit, which shall be an assembly of one rotary switch with its attached wiring, one switch control shaft with a plastic control dial mounted thereon, three fuse holders each containing one fuse, and one AN receptable.

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b. One Cable Assembly, Special Purpose, Electrical CX-4701/GWM-8A, 8 feet in length, Air Force drawing 59C45037. This cable assembly shall have one connector at one end to connect to the external receptacle on the RYK-2/A24T-1 or to receptacle "A" of the bomb bay control and monitor junction box. There shall be one connector at the opposite end of the cable assembly to connect to the receptacle on the cylindrical shape of the testing unit.

3.4 Design and construction. The design and construction of the AN/GWM-8A Test Set shall be in accordance with drawing 59D45036.

3.5 Performance. The AN/GWM-8A Test Set shall be capable of functional operation that will insure satisfactory performance of the Tester, In-Flight Control T-249 or the AF/A24T-1 Bomber Aircraft Control Monitor System, and the wiring systems used therewith.

* 3.5.1 The AN/GWM-8A Test Set shall be capable of satisfactory performance as specified in 3.5 after being subjected to the following servicie conditions:

a. Vibration. Vibration incident to service use.

b. Shock. Shock forces incident to service use.

c. High temperature. Temperatures up to and including 71° Centigrade (C) (160° Fahrenheit (F)). The temperature may remain constant or vary at a rate as high as 1°C (1.8°F) per second.

d. Low temperature. A low temperature of -57°C (-70°F). The temperature may remain constant or vary at a rate as high as 1°C (1.8°F) per second.

e. Humidity. Relative humidity up to 94 ±4 percent including conditions wherein condensation takes place in the form of both water and frost.

f. Salt spray. Exposure to salt sea atmosphere.

g. Sand and dust. Sand and dust as will be encountered in service in desert areas.

h. Fungus. In accordance with method 508.1 of MIL-STD-810.

3.6 Identification of product. Equipment assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

* 3.7 Reference designations. Reference designations used for markings within the unit shall be in accordance with ANSI Y32.16 markings shall not adversely affect the leakage path between conductors, or any other factor of performance. Reference sysmbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily viewed.

3.8 Workmanship. The AN/GWM-8A Test Set, including all prts and accessories, shall be fabricated and finished in a workmanlike manner to produce units free from all defects. Particular attention shall be given to neatness and

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thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test set shall be as follows:

a. Preproduction tests

b. Acceptance tests

4.3 Preproduction testing

4.3.1 Preproduction test samples. The preproduction test samples shall consist of model representative of the production items. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under supervision of the procuring activity.

* 4.3.1.1 In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report and funish three complete copies of the report to the procuring activity.

4.3.2 Preproduction tests. Preproduction tests of the AN/GWM-8A Test Set shall consist of all tests described herein under 4.6.

4.4 Acceptance tests. Each test shall consist of the following:

a. Individual tests

b. Sampling plan and tests

4.4.1 Individual tests. Each test set shall be subjected to the following tests as described under 4.6 of this specification:

a. Examination of product

b. Continuity test

4.4.2 Sampling plan and tests. Sampling plan and tests for the AN/GWM-8A Test Sets shall be in accordance with 4.4.2.1 of this specification. No sampling plan for the wiring harness shall be required.

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4.4.2.1 Periodic sampling plan. AN/GWM-8A Test Sets shall be selected at random in accordance with the following schedule, and shall be subjected to the tests listed below and described under 4.6:

SAMPLING SCHEDULE

Consecutively Produced Units on Order	Number of Units to be Sample Tested
99 or less	2
100 to 299	1
Each 200 or fraction thereof on balance of order	1 unit for each additional 200 produced

a. Vibration

b. Shock test

4.4.2.1.1 Rejection and retest. When one or more items from a production run fail to meet the specified test, no items still on hand or later produced shall be accepted until the extent and cause failure are determined.

4.4.2.1.2 Individual test may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of this specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25°C ±10°C and a relative humidity of 80 percent or less. Where tests are made at atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.6 Test methods.

4.6.1 Examination of product. Each AN/GWM-8A Test Set shall be examined to determine compliance with requirements of this specification, including materials, workmanship, ease of operation of control, and strict conformance to the pertinent drawings.

4.6.2 Continuity test. Continuity test shall be conducted on each AN/GWM-8A Test Set to insure continuity of all electrical circuits, in accordance with 4.6.2.3 of this specification.

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4.6.2.1 Test circuit. The acceptance tester for the AN/GWM-8A Test Set shall be any reliable ohmmeter.

4.6.2.2 AN/GWM-8A schematic. The complete wiring schematic for the AN/GWM-8A and its wiring harness is shown on Air Force drawing 59C45039.

4.6.2.3 Test procedures. Each series loop of the tester and cable assembly, as listed in Table I of this specification, shall be checked for continuity. The sequence of the rotary switch test positions may be rearranged for the convenience of person or persons performing the testing.

TABLE I. Continuity test for CX-4701/GWM-8A cable assembly.

Position	Continuity
1	G(P1) - a, Z Y(P1); K, h(P1) - c(P1); L(P1) - O(P1)
2	G(P1) - d, B, D(P1); K, h(P1) - T(P1); L(P1) - O(P1)
3	G(P1) - U(P1); L(P1) - A(P1)
4	G(P1) - f(P1); K, h(P1) - F(P1); L(P1) - O(P1)
5	G(P1) - c(P1); K, h(P1) - k, W(P1); L(P1) - S(P1)
6	G(P1) - g(P1); K, h(P1) - b(P1); L(P1) - U(P1)
7	G(P1) - f(P1); K, h)P1) - A(P1); L(P1) - V(P1)
8	G(P1) - a, Z, Y(P1); K, h(P1) - J(P1); L(P1) - V(P1)
9	G(P1) - f(P1); K, h(P1) - g(P1); L(P1) - V(P1)
10	G(P1) - W, k(P1); K, h(P1) - c(P1); L(P1) - V(P1)
11	G(P1) - T(P1); K, h(P1) - A(P1); L(P1) - V(P1)
Any Position	(P1) a, Z, Y; (P1) B, D, d; (P1) W, k; (P1) E, C, R; (P1) H, P; (P1) h, K

4.6.2.4 Failure to obtain the prescribed indication during any step of the test procedures outlined in 4.6.2.3 shall be cause for rejection of the AN/GWM-8A Test Set being tested.

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4.6.3 Environmental tests.

4.6.3.1 Vibration test. The vibration test shall be in accordance with vibration tests, method 513.2, procedure X of MIL-STD-810. At the end of the vibration tests, the test specimen shall be inspected for ease of operation of controls, and tested in accordance with 4.6.2.

* 4.6.3.2 Shock test. The AN/GWM-8A shall be dropped 3 times from a height of 42 inches. The AN/GWM-8A shall strike the impact surface on its cylindrical surface. The impact surface shall be 3/4 inch plywood backed by concrete. After the three impacts, the AN/GWM-8A shall be inspected for ease of operation of controls, and tested in accordance with 4.6.2.

* 4.6.3.3 High temperature test. The high temperature test shall be in accordance with the high temperature test, method 501.1, procedure II of MIL-STD-810. Exceptions to MIL-STD-810 are: At step 7 reduce the temperature to ambient temperature and inspect the test item for ease of operation of controls and test per 4.6.2. Omit steps 8, 9 and 10 of MIL-STD-810.

* 4.6.3.4 Low temperature test. The low temperature test shall be in accordance with the low temperature test method 502.1, procedure I of MIL-STD-810. Omit steps 3, 4 and 5. In step 7 inspect the test item for ease of operation of controls and test per 4.6.2 in lieu of that stated in MIL-STD-810.

* 4.6.3.5 Humidity. The humidity test shall be in accordance with that required by method 507.1, procedure II of MIL-STD-810. Initial tests measurements and post-test measurements are those per 4.6.2. Test item is not required to operate during step 6. In step 9 test the item to 4.6.2 in lieu of that called for in MIL-STD-810.

* 4.6.3.6 Salt spray. The salt spray test shall be in accordance with method 509.1, procedure I of MIL-STD-810. Pre-test and post-test criteria are inspection for ease of control operation and test per 4.6.2.

* 4.6.3.7 Sand and dust. The sand and dust test shall be per method 510.1, procedure I of MIL-STD-810. Pre-test and post-test criteria are (a) inspection for ease of control operation and (b) test per 4.6.2.

* 4.6.3.8 Fungus test. The fungus test shall be conducted per method 508.1, procedure I of MIL-STD-810. Pre-test data are not required. Post test shall be an inspection to determine fungus growth inside and out, operation of operating controls and test per 4.6.2.

4.6.3.9 Cause for rejection.

4.6.3.9.1 Failure to operate properly or easily, or failure to pass the tests specified in 4.6.2 following any of the environmental tests, shall constitute failure to pass the particular environmental test.

4.6.3.9.2 Damage and deterioration, or corrosion of any internal or external component, which could in any manner prevent the equipment from meeting operational requirements during service life, shall provide reason to consider the equipment as having failed to meet the test to which it was subjected.

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5. PACKAGING

5.1 Preservation, packing, and packaging. Preservation, packing, and packaging for shipment shall be in accordance with MIL-P-17555.

5.2 Marking for shipment and storage. Preservation, packing, and packaging for shipment shall be in accordance with MIL-P-17555. The shipment marking nomenclature shall be:

Test Set, Armament Wiring System, Special Weapon AN/GWM-8A

6. NOTES

6.1 Intended use. The AN/GWM-8A Test Set is intended to be used for check-out of the Tester, In-Flight Control T-249 and bomber aircraft wiring used therewith.

6.2 Ordering data. Invitations for bids, contracts, and purchase orders should state the conditions for the following:

6.2.1 Preproduction samples. It is expected that the contract or purchase order will specify that two units representative of the production test set will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should state the point of inspection for these tests.

6.3 Sources. When requesting publications, refer to both title and number.

6.4 Definitions.

a. Preproduction tests. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order to prove, prior to his starting production, that the test set meets the requirements of this specification. Any production prior to the approval of the preproduction samples shall be at the contractor's risk. See 4.3.

b. Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification and to determine that the units are equivalent in performance and construction to the approved preproduction samples. See 4.4.

* 6.5 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible. Reclaimed materiel shall be limited to raw materials and shall not include surplus components, switches, et cetera.

6.6 Changes from previous issues. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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